Design a URL Shortener (Bit.ly)

Q. How many people we are targeting ?
A. 20 million URLS in a day

Q. (www.example.com/sps/1/pwm-18uuaName=Aditya

www.bit.ly/qOw12

Q. How small the short URL?
A. As short as possible.

Q. What characters i can use to create Short URL?

A. (0-9, a-z, A-Z)

FLOW:
1. user comes to our service by giving us a long URL.
2. we will give him back a short URL.
3. User put this short URL in browser & is redirected to the long URL.

Z base
short URL generate
user Redirect

Back of the envelope calculations:

QPS (Query per second):
20 million URLs generated per day

Write Operations : 20 million / day
Write operation/ sec:
20 * 2^20 / 86400 sec
= 20 * 10^6 / 86400 sec
= 231.48 * 2{ -20 } 10^(-6)
231 * 2^-20 * 2 * 20
~ 231 operations/sec

Read Operations:
1 : 10 ratio read:
Read Operations/sec = 231 * 10
= 2310 operations/sec.

Storage Unit:

Assume for 5 years.
20 million records / day
20 * 2^20 * 365 days * 5 years records
= 20 * 10^6 * 365 * 5 records
= 36.5 * 10^6 * 5 * 20 records
= 36.5 * 2^10 * 5 * 20 records
= 36.5 Billion records to store

Assume 1 record = 100 Byte.
36.5 Billion * 100 bytes
= 36.5 * 2^30 (1024 * 2^10) 10^3
= 3.65 * 2^40
= ~ 3.65 TB

API end points.
1. Generate a short URL
2. Redirect to long URL

POST
GET

These appear after ? in the URL
https://www.bit.ly/api/v1/generate?longURL=https://google.com

Generate a short URL:
POST : www.bit.ly/api/v1/generate

OR

longURL : value

query parameter

OR

POST : www.bit.ly/api/v1/generate/{longURL}

worst

OR

POST : www.bit.ly/api/v1/generate
Body {
longURL : value
}

1. Generate a short URL:
POST : www.bit.ly/api/v1/generate
Body {longURL}

Return : shortURL

2. Redirect to Long URL:
GET : www.bit.ly/{shortURL}
Response : Redirect to Long URL

How to Redirect?

Status Code :

301 : Redirection : Permanent Redirection

302: Redirection : temporary redirection

user do visit our app since he gets the url from cache
hence minimum revenue
no user metrics

How do we decide ?
301: bit.ly/QW123 → {long URL}
Server is small
DB Read Query Minimum.

302 : bit.ly/QW123 -> Service -> {longURL}
server is scalable
DB Read Query is optimized.
We can have user metrics.

use 302 for your application.

Design an Initial Flow

1. URL Shortening (POST)
2. URL Redirect (GET)

1. URL Shortening

POST : www.bit.ly/api/v1/generate
Body {longURL}
Return : short URL

Mapping : Long URL -> short URL

Long URL -> mapped to -> short URL

Hash Map / hash table
table.get(shortURL)
= longURL

My - Short URL
Value - Long URL

How to generate short URL from LONG URL ??

Hash Functions

LONG URL -> Hash function -> SHORT URL

because hash functions returns unique value

Storing:
1. user comes with a long URL
2. We generate a short URL for the same
2. Hashmap.put(key, value)
3. Store the shortURL as key and
LongURL as value in our hashmap.
4. Return the Short URL

Redirecting:
1. User comes with a short URL.
2. We find the equivalent Long URL by doing
map.get(shortURL).
3. Redirect the user to long URL (302 redirect)

DB : HashMap -> NoSQL (Amazon Dynamo) single key single value

MongoDB (complicated) single key but one or more values

Amazon S3

HLD : Distributed Applications

Multiple web servers
Multiple DB servers (Db Replication and Scaling)

Sharding (horizontal scaling)

Consistent Hashing

Now we have 2 Hash Functions:
1. Find out the Short. (DB: DHA-1)
2. Find out the shortURL from longURL (MD-5)

Better Design

NoSQL -> SQL DB

Why ??

1. long URL -> hash function -> short URL
2. long URL -> hash function -> short URL 2nd write operation with same long url

Dynamo DB

key	Value
short URL	Long URL

but in key value pair you cannot query on key but you can query on values in order of W1. if you are querying values then it is 50x which is worst

2. Production level APP -> for a particular user how many URLS are generated till now.

SQL -> support Complex Queries efficiently

3. write operations are faster in SQL than in NOSQL

SQL Table : Indexing

ID (PK) shortURL LongURL
primary key

Hash Functions ??

Convert a long url to a short URL

MD5, SHA-1, SHA-256 etc hash value is too long for this hashing techniques

Hash value length :
Possible symbols : (0-9, a-z, A-Z)

Total = 10 + 26 + 26 = 62 characters

Interviewer : — keep it as small as possible. (How small)
www.bit.ly/qwd (min characters)

Using Permutations:
www.bit.ly/_ _

If we take just 1 character -> 62 possible URLS
If we take just 2 characters -> 62^2
If we take 3 -> 62^3.

How much char i should take -> n -> 62^n possible URLS

36.5 Billion records to store

62^n <= 36.5 Billion

n = 6 (56 Billion Records)

www.bit.ly/_ _ _ _ _ _

N = 6.

SHA-1 , MD5, SHA256, UUID generates long hash

aafce34 223ecpee234132ecpe23
1

if you want to trim the values of the existing algorithms

1. Hashing using existing Algorithms
Will lead to too much collisions.

HASH + COLLISION REMOVAL

Disadvantage:
1. DB lots increases.
2. Slow. (Recursive calls).

BASE X Approach base 10 -> base 2
2 -> 10
3 -> 11

62 possible characters

Number (Base 10) -> Base 62

62 Numbers (0-9, a-z, A-Z)
0 - 0
1 - 1
2 - 2
10 - a
15 - b
35 - z
36 - A
61 - Z

Convert 14320 into Base 62:

SQ

id	shortURL	longURL
14320	3Y	————

since id is unique for every record there will be no collision because hash function values for different ids.

no hashing of long url

base 62 of id

Long URL : www.example.com
ShortURL : www.bit.ly/3Y

Advantage:
1. Collision is impossible. this advantage overpowers its disadvantages

Disadvantage:
1. Need a unique ID generator
2. URL length is not fixed. but never the exceeds the length (as it is this case)
3. It is easy to figure out the next generated
short URL (if id is incremented by 1)

Example:
Long URL :
https://www.example.com/get/code/saved=124

POST

Unique ID : 14320
Short URL : base62(14320) = 3Y

SQ

id	shortURL	longURL
14320	3Y	————

We need to build a unique ID generator

Distributed System

Multiple DBs (shards) :- Each will generate there own IDs. 2 IDs can be same. hence we cannot use IDs from DBs we need unique ID generator

Centralized ID generator

Storing

Redirection

Distributed URL Shortener

What if unique ID generator generates a value that when converted to base 62 is more than 6 digit ?

Not possible

62^6 = 56 Billion unique records
36.5 Billion to be stored actually

62^7 = 3.5 Trillion